SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Classify various types of process control systems. Explain the working of feed forward & feed backward control system.

Q.24 Explain with neat diagram the working and constructional details of radiation thermal detector (RTD) pyrometer.

Q.25 Write short notes on any two of the following:

a) Orsat analyzers

b) Viscosity measurement

c) Instrumental and environmental errors.

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Roll No. ....

6th Sem / Chemical (Pulp & Paper)

Subject : Process Instrumentation & Control

Time : 3 Hrs. M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Optical level detector uses

a) Sound b) Gamma rays

c) Light d) none

Q.2 In pressure level gauge indicator, pressure gauge is mounted at

a) top of the tank

b) lowest level of the tank

c) both at top & bottom

d) none

Q.3 The range of air purge system is determined by the

a) air pressure b) tank height

c) Length of tube d) None of the above

Q.4 ______ one the Fahrenheit Scale, the interval between lowest and upper fixed point is divided into

a) 180 equal parts b) 100 equal parts

c) 90 equal parts d) 50 equal parts

Q.5 Which device is used for calibration pressure gauges:

a) manometer

b) diaphragm

c) bellows

d) Dead weight pressure tester

Q.6 Radiation level detector is

a) contact device b) contact less device

c) Both a & b d) none

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Define precision.

Q.8 Define dead zone.

Q.9 Define span.

Q.10 Define static error.

Q.11 Define range & zero

Q.12 Define PH.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Explain capacitance level indicator with neat transfer.

Q.14 Explain pressure gauge with neat diagram.

Q.15 Explain with neat diagram the working principle of bimetallic thermometer.

Q.16 Explain in brief method gauge used for measurement of vacuum

Q.17 Explain oxygen analyzers in brief.

Q.18 Explain PH meter & its applications.

Q.19 Explain closed loop (Feedback control system with neat diagram.

Q.20 Write short notes on controller variable and manipulated variable.

Q.21 Explain valve actuator and valve positioning.

Q.22 Explain ultrasonic liquid level detector with neat diagram.